TeleCon4ESP32 User Guide

Phone app for Control Panel and RC Vehicle

For people who use the Android application

Firmware authors: see the Control Protocol PDF

Version 1.0 2026

Abstract

This guide is for the person holding the phone. It shows how to install TeleCon4ESP32, flash a starter ESP32 sketch, connect, and drive **Control panel** or **RC Vehicle**. Byte formats, UUIDs, and checksums live in the Control Protocol document. Per-sketch pinouts and Serial banners live in that sketch's user guide.

1 What TeleCon is

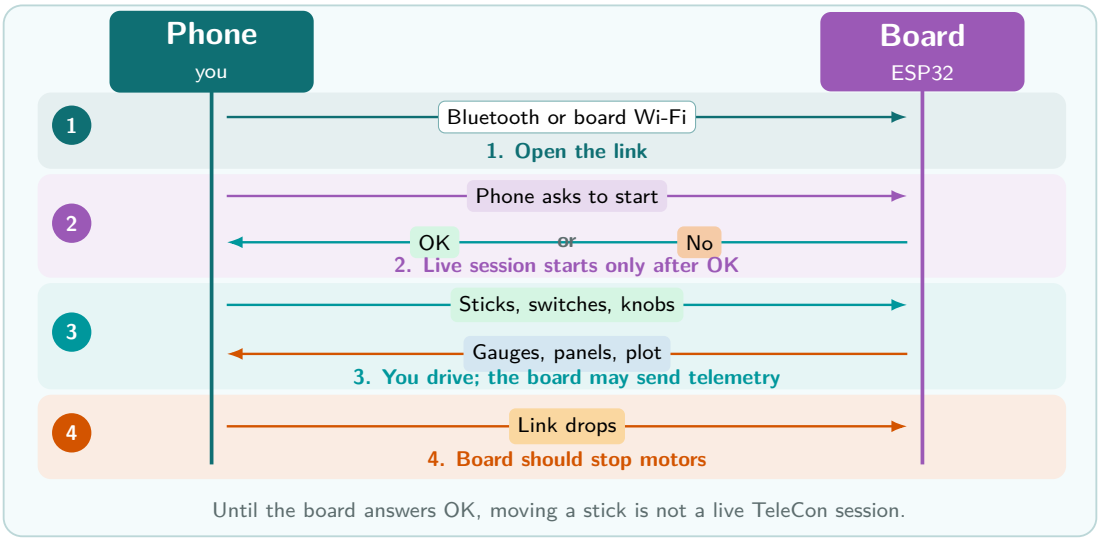
The phone is always the remote. The board (ESP32 DevKit, ESP32-CAM, or any MCU that speaks the contract) is the machine.

Two modules ship today:

Module	Access	What you do with it
Control panel	FREE	Dual sticks, switches, knobs, gauges, and plots for any ESP32 project.
RC Vehicle	PRO / coins	Driving cockpit plus optional ESP32-CAM video.

Control panel — free RC Vehicle — Pro Video is a separate Wi-Fi link

Both modules share the same control session. Camera video (HTTP `/stream`) is a second link: join the board's Wi-Fi for pictures; Bluetooth or TCP carries sticks.



2 What you need

- An Android phone (Android 12 or newer is simplest for Bluetooth).
- USB cable and a computer with Arduino IDE (to load the starter sketch).
- An ESP32 DevKit for the first session. An ESP32-CAM is optional and only needed for video.

Start with **Classic Simple** until that path works. Classic Binary and BLE Binary are in the same connection list.

3 Safety

- Treat sticks as live once the app shows **Connected**. Wheels, servos, and ESCs can move.
- Until the board answers OK, on-screen motion is not a live session even if Android says Bluetooth is connected.
- If the phone drops the link, official sketches fail-safe (motors stop). Confirm that on your own firmware before a vehicle leaves the bench.
- One phone at a time. Close other apps that hold the same Bluetooth or Wi-Fi link.

4 First session

Goal: in about ten minutes, move a stick on **Control panel** and see the board react (Serial Monitor echo, or a motor if you already wired one).

Before you start. Phone charged. ESP32 powered from USB. Arduino IDE installed. You do not need Pro or coins for this path.

1 — Install the app

Install TeleCon4ESP32 from Google Play (or your debug build). Open it. Grant Nearby devices / Bluetooth when Android asks. On Android 11 and below, Location permission is still required for scanning; the app does not need Location to stay on.

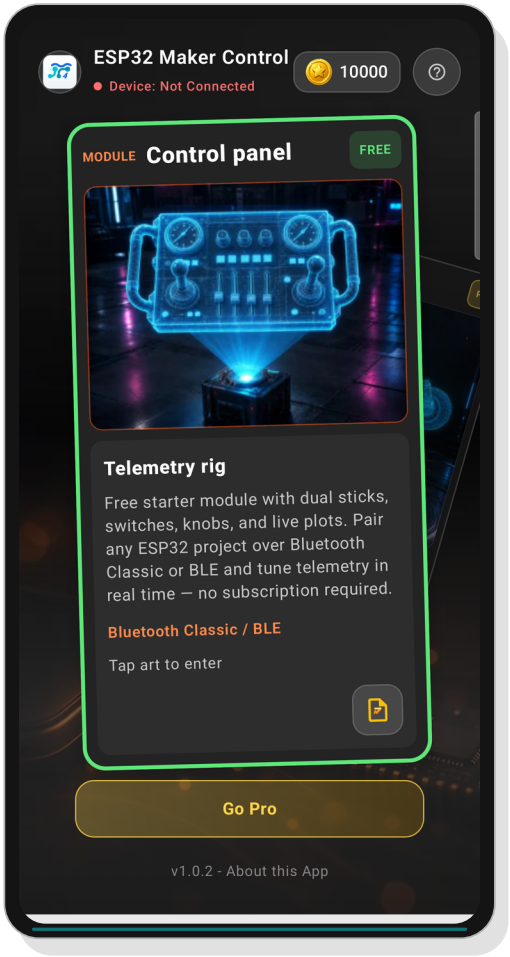


Figure 1: **S01.** Home screen after first launch. Show the top bar (connection status Not connected, optional gold PRO, coin chip, Help), the two module cards Control panel (FREE) and RC Vehicle (PRO), and the Codes icon on a card. Do not cover labels. English UI.

Portrait, light theme, not connected ▪ `s01_home.png`

2 — Open the guides

On Home, tap the docs icon on **Control panel**. **Codes and Documents** shows the **Fast guide** and **User guide** PDFs. Get the starter sketch ZIP from **Matching firmware** in settings (next step).



Figure 2: **S02.** Codes and Documents for Control panel. Fast guide and User guide only. English UI.

Portrait, light theme ▪ `s02_codes.png`

3 — Get the starter sketch, match settings, then connect

Open **Control panel** → settings (gear). **Connection** = Bluetooth, **Classic + Simple (text lines)**. Under **Matching firmware**, tap the matching code link for `TeleCon_Classic_Simple`, share or save the ZIP, copy it to the computer, and open the `.ino` folder in Arduino IDE.



Figure 3: **S03.** Control panel settings. Connection Bluetooth + Classic + Simple (text lines), and Matching firmware with the TeleCon Classic Simple code link. English UI.

Portrait, light theme, Connection visible ▪ `s03_settings_default.png`

Flash **TeleCon Classic Simple**. Board and Port under Tools must match your USB device. Serial Monitor at 115200. Press RESET. Wait for the ready banner, then the board is advertising `ESP32-TC-RC-BT-Simple`.

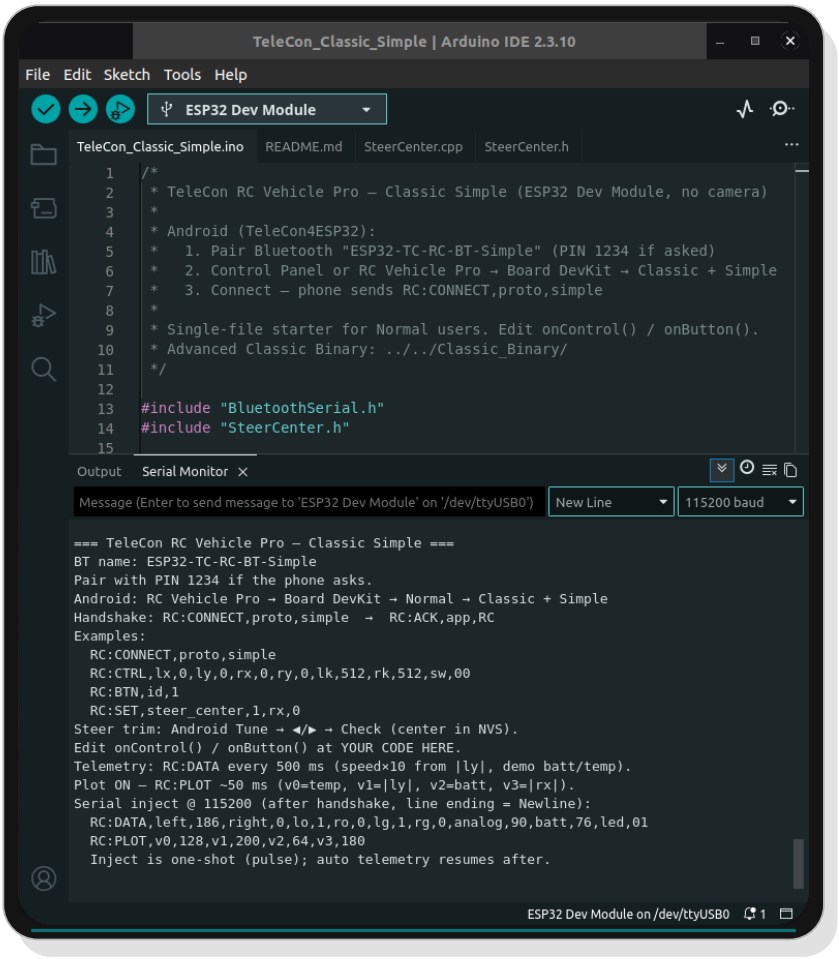


Figure 4: **S02b.** Serial Monitor at 115200 after RESET. Classic Simple ready banner and BT name `ESP32-TC-RC-BT-Simple` visible. Crop to the Serial window.

Computer, Arduino IDE Serial Monitor ▪ `s01_serial.png`

Pinouts and motor hooks stay in that sketch's `README.md` next to the `.ino`.
Leave settings, then tap the Bluetooth button on the module toolbar.
If Android asks to turn Bluetooth on, allow it.

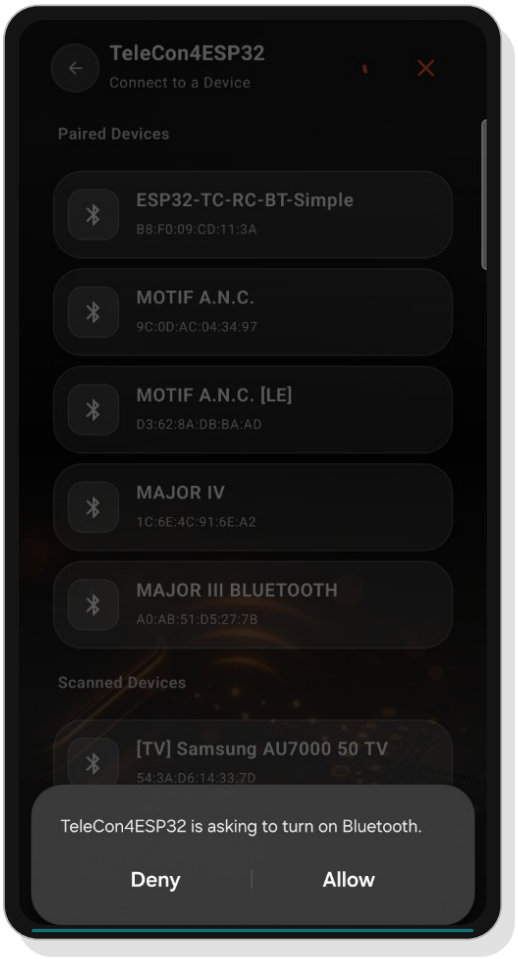


Figure 5: **S04.** Android Bluetooth enable sheet from the Connect screen (Deny / Allow). English UI on this capture.

System dialog, portrait ▪ `s04_permission.png`

On **Connect to a Device**, tap refresh to scan. When `ESP32-TC-RC-BT-Simple` appears under **Scanned Devices**, stop the scan and tap that row. If it was paired before, use **Paired Devices**.



Figure 6: **S05.** Bluetooth screen with Paired Devices and Scanned Devices. ESP32-TC-RC-BT-Simple visible. English UI on this capture.

Portrait, Connect screen ▪ `s05_bluetooth_scan.png`

Expected result. The app returns to the module. Status shows **Connected**. Home also shows Connected and highlights Control panel.

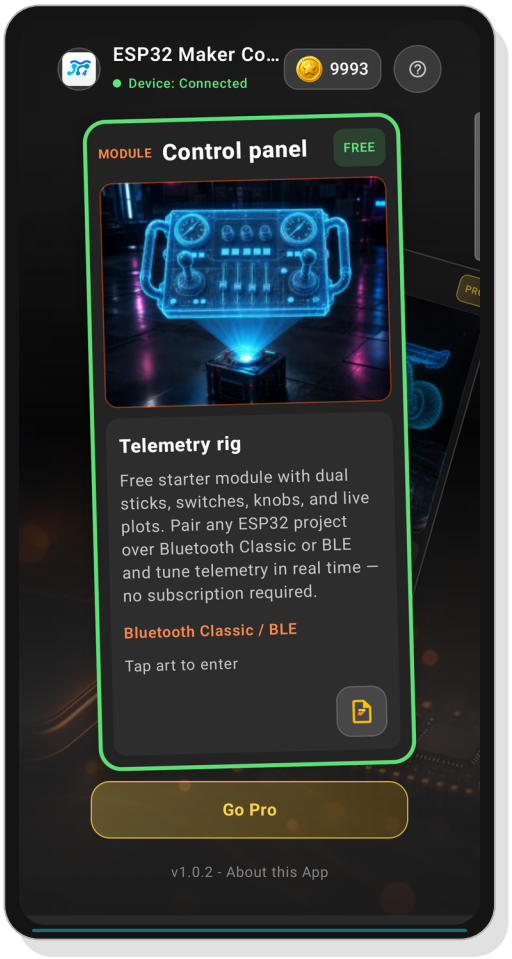


Figure 7: **S06.** Same Home layout as S01, but the top status reads Connected and the Control panel card is the highlighted recent module.

Portrait, light theme, connected ▪ `s06_home_connected.png`

4 — Open Control panel and move a stick

Open **Control panel** (landscape). Drag the left stick. On the bench, official Simple firmware echoes motion into gauges and the plot (simulate mode, no extra wiring). In Serial Monitor you should see control traffic.



Figure 8: **S07.** Full Control panel HUD: both sticks, S1–S6, both knobs, side buttons, header gauges, seven-segment panels, LEDs, and the center plot. Connected (no "Not connected" banner). Optional: left stick off-center so the layout is obvious.

Landscape, light or the app's plastic theme ▪ `s07_control_panel.png`

If nothing moves, stay on this screen, confirm Connected, and check that settings still say Classic + Simple — see Section 12.

5 Home

Home is the launcher, not the live controller.

Control	What it does
Control panel / **RC Vehicle**	Opens that module. Expand a card for details, docs, or coin unlock.
Docs icon on a card	User guide and Fast guide PDFs for that module (sketch ZIPs are in Settings → Matching firmware).
Gold **PRO**	Subscription / upgrade.
Coin chip	Balance and pricing (when the coin economy is on).
Help (question mark)	Short FAQ for Home.
Title **TeleCon4ESP32**	About, privacy, and online documentation.

Browse Catalog appears only when more modules ship. Until then, tap a card on Home.

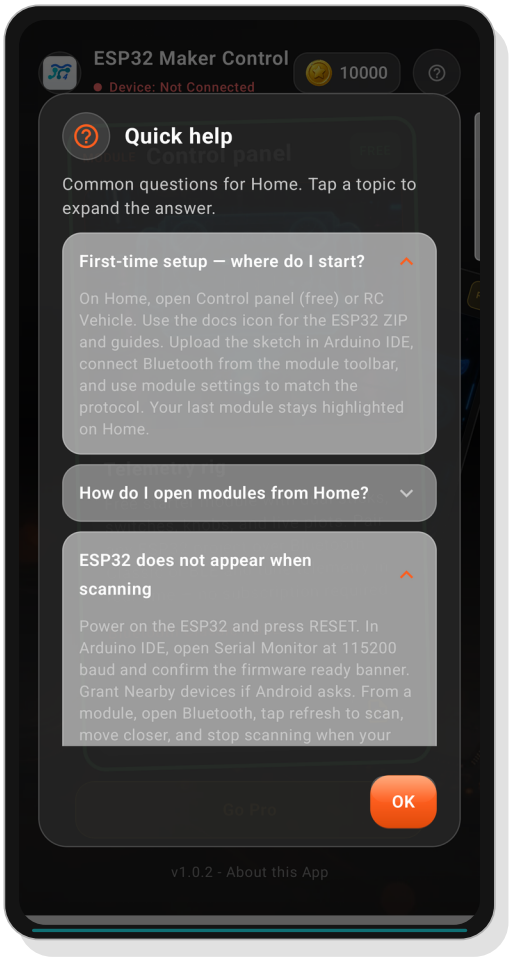


Figure 9: **S08.** Home with the Quick help dialog. Title Quick help, at least two topics expanded (first-time setup and ESP32 does not appear). English UI.

Portrait, Help dialog open ▪ `s08_home_help.png`

6 Connect

Always connect *from the module* you will use (toolbar Bluetooth / Connect), not from a leftover system pairing alone. The module settings tell the app which link to open.

6.1 Bluetooth

Used by DevKit sketches: Classic Simple, Classic Binary, and BLE Binary.

1. Allow Nearby devices (or Location on older Android).
2. Scan, then pick the board. The advertised name is only a label; the app connects by address.
3. Official Default name: `ESP32-TC-RC-BT-Simple`.

If a previous firmware is still bonded, Android may open the wrong profile. Forget the device in system Bluetooth settings and scan again.

6.2 Board Wi-Fi (SoftAP)

Used by the video-only ESP32-CAM. The phone leaves home Wi-Fi for the picture. No internet on that network is normal. Sticks stay on Bluetooth.

Camera network: SSID `TeleCon-RC-CAM-Starter`, password `telecon1234`. The camera board does not take stick control.



Figure 10: **S09.** System Wi-Fi picker with TeleCon-RC-CAM-Starter visible under Available networks (On). English system UI on this capture.

Portrait, Android Wi-Fi settings ▪ `s09_wifi_join.png`

Do not put SoftAP and Bluetooth on a single ESP32-CAM. Video CAM + control DevKit is the two-board (Role A) layout.

7 Control panel

Landscape remote for any ESP32 project. While this screen is open, sticks and knobs stream continuously.

7.1 Layout

Numbers match the callouts you will add on screenshot S07:

1. Header gauges — analog value and battery-style level from telemetry.
2. Seven-segment panels — large numbers (for example RPM). Titles and color come from Panel settings; values come from the board.
3. Status LEDs — eight lamps driven by a telemetry bitfield.
4. Joysticks — left and right. Spring vs hold is a setting.
5. Switches S1–S6 — three per side.
6. Rotary knobs — auxiliary 0–100% style values.
7. Action buttons — four momentary side buttons.
8. Center area — plot by Default; Camera, Radar, and Stick graph are extra unlocks.

7.2 Panel settings

From the module, open settings → initial switches, initial knobs, panel labels, plot channel names.

On the live Control panel HUD, **double-tap the stick** you want to configure (left or right). That opens the joystick sheet for that stick (Spring / Hold, axis orientation, stick shape).



Figure 11: **S10.** Control panel live HUD after double-tapping a stick: joystick settings sheet open (Spring / Hold, axis orientation, stick shape). English UI on this capture.

Landscape, Joystick Settings visible ▪ s10_panel_settings.png

Center display (Plots / Stick / Camera / Radar) is a circular picker on the live HUD. Camera and Radar need coins or Pro.



Figure 12: **S11.** Control panel with the Center display sheet: Plots, Stick, Camera, Radar. Show which items are locked if running a free account.

Landscape, center-display picker open ▪ `s11_center_picker.png`

8 RC Vehicle

Pro driving cockpit. Unlock with subscription, coins, or the occasional explorer gift (short ad).

One hardware layout for video:

Layout	Boards	What the phone does
Camera overlay	CAM + DevKit	CAM Wi-Fi for video only. Sticks on DevKit Bluetooth.

CAM video + DevKit Bluetooth

Steer center trim lives under Tune on the HUD (tickers → Check). The board stores the bias.

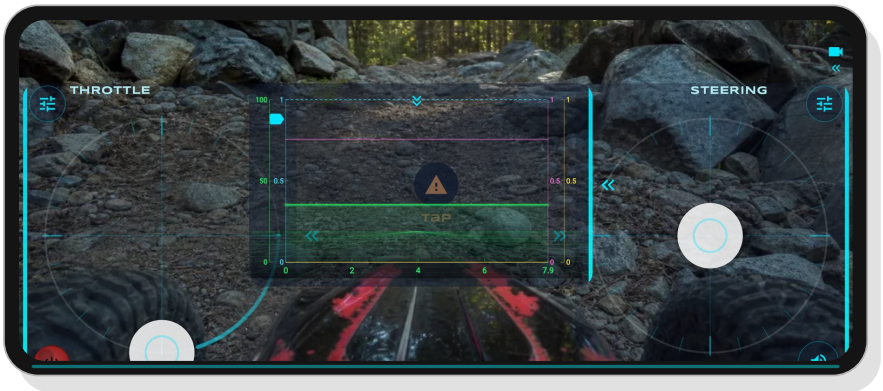


Figure 13: **S12.** RC Vehicle live screen: dual sticks, HUD telemetry, and the camera pane (live MJPEG or the in-app camera help card if the stream is idle). English UI. Video, when used, comes from a separate ESP32-CAM.

Landscape HUD ▪ `s12_rc_vehicle.jpg`

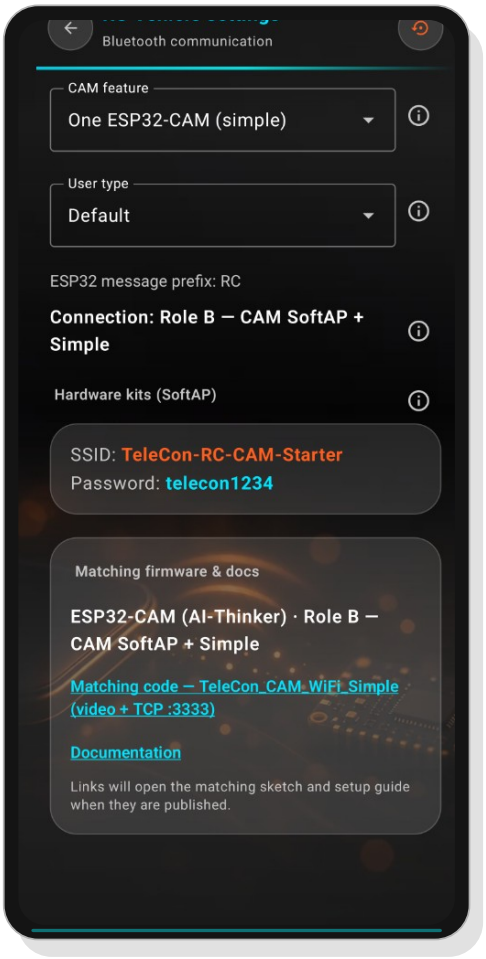


Figure 14: **S13.** RC Vehicle settings: optional SoftAP camera on a second ESP32-CAM, SSID TeleCon-RC-CAM-Starter. Sticks stay on DevKit Bluetooth. English UI on this capture.

Portrait, Connection / CAM section ▪ `s13_vehicle_settings.png`

9 Connection modes

Classic Simple, Classic Binary, and BLE Binary are all listed in module settings. Pick the row that matches the sketch.

Mode	Link
Classic Simple	DevKit text lines, or the same mode plus a video-only ESP32-CAM.
Classic Binary and BLE Binary	DevKit binary frames. The same optional video-only ESP32-CAM.

The firmware on the board must match the row you pick. After a change, disconnect and connect again.

10 Codes and About

- **Codes and Documents** — this User guide PDF and the short Fast guide for the module.
- Module **Settings** → **Matching firmware** — share or save the sketch ZIP that matches the board + connection you selected.
- **About** — version, privacy policy, and the hosted documentation site.

11 Coins and Pro

Control panel stays free on Classic Simple. **RC Vehicle** and extra center views (Camera, Radar, Stick graph, session CSV) need coins or a Pro subscription.

Where to look. The coin chip in the Home top bar shows your balance. Tap it to open **Coin pricing** (wallet and rate card). On a locked Pro module card, use the unlock / coin actions, or **Go Pro** at the bottom of Home for a subscription that removes the coin/ad path.

Earn coins. Watch a rewarded ad from the pricing sheet (play button, typically +3 coins per ad). Ads may be unavailable sometimes; try again later.

Spend coins. Coins unlock **one Pro module at a time** for a chosen duration. Typical options (exact costs are on the sheet in the app):

Option	Meaning
One use	Single session-style unlock of that module.
24 hours	Timed access for one day.
3 days / 1 week	Longer timed access; usually better cost per day.

Open the unlock dialog on the module you want, pick a duration, and confirm if your balance is high enough. Longer packs usually lower the effective coins-per-day shown in the pricing table.

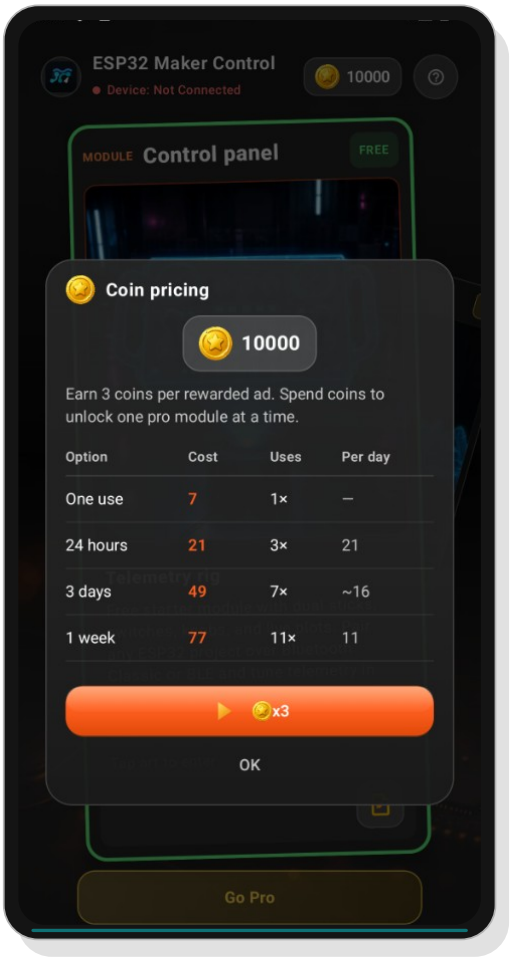


Figure 15: **S15.** Home coin wallet / Coin pricing dialog: earn rate (+3 per rewarded ad), spend line, and the Option / Cost / Uses / Per day table. English UI.

Portrait, Coin pricing sheet ▪ `s15_coin_pricing.png`

Explorer gift. The gold sparkle on a locked card is an occasional explorer gift: watch a short ad to unlock that Pro module for a few hours (about four). The sparkle returns after a few days.

Pro subscription. **Go Pro** unlocks Pro modules without spending coins and without the rewarded-ad earn loop. Details and pricing are in the Play subscription sheet.

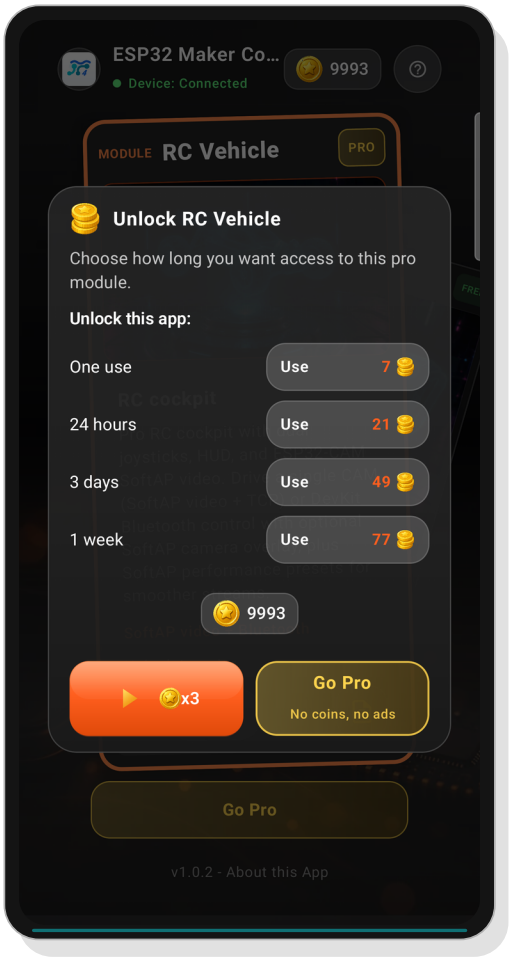


Figure 16: **S14.** Coin unlock dialog for a specific module (for example RC Vehicle): duration choices and confirm. English UI.

Portrait, unlock dialog ▪ `s14_coin_unlock.png`

12 When it fails

Start from the symptom, not from the protocol.

You see	Try this
Board missing in scan	Power + RESET. Serial ready banner at 115200. Nearby devices allowed. Move closer. Stop scan when it appears.
Connect fails	Open Bluetooth from the module, not only from system settings. Forget a stale pair. Match Classic vs BLE to the sketch.
Home still Not connected	Connect inside the module. One phone only. Then Home updates.
Sticks / gauges dead	Stay on the live screen. Settings must match the sketch (Classic Simple for the first session). Reflash if Serial never shows ready.
"Couldn't connect" / handshake timeout	The phone opened a pipe but the board did not answer OK in time. Same mode on both sides; update the sketch if it ignores the start request.
Protocol doesn't match	Settings say Simple and the board speaks Binary (or the reverse). Change settings or flash the matching ZIP.
No camera	Join the CAM SSID first. Password `telecon1234`. Role A: video on CAM Wi-Fi, sticks on DevKit Bluetooth — not both radios on one CAM.
Motors keep running after drop	Official sketches stop. If you wrote firmware, add fail-safe before any vehicle test.

Xiaomi / POCO on SoftAP: turn off mobile data if the TCP link dies, and keep TeleCon in front. VPN off.

13 Where to go next

- Sketch **README** next to the `.ino` — pins, SSID, BT name, `YOUR CODE HERE`.
- **Control Protocol** PDF — if you write your own firmware.
- In-app Help — short reminders on the phone.